Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Universal Physics-Informed Neural Network Platform for Multiphysics Simulation and Digital Twin Applications*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Software Solutions*	*Software*	*Internet SW*
Concept of Proposed Technology	The proposed technology is a Universal Physics-Informed Neural Network (PINN)–based software platform designed to replace or augment conventional multiphysics computer simulations. The core concept is to embed governing physical laws directly into neural networks, enabling fast, reliable, and reusable solutions for complex coupled multiphysics problems that traditionally require high computational cost and long simulation times. ***Core Concept*** Conventional multiphysics simulations (e.g., structural dynamics, CFD, thermal-fluid interaction, material degradation) require full numerical re-solving whenever boundary conditions, geometry, or material properties change, resulting in excessive computational overhead. The proposed technology addresses this limitation by introducing a modular, plug-in–based PINN architecture in which physics models, boundary conditions, constitutive laws, and coupling strategies are encapsulated as reusable			

software modules [1]. Once a base PINN model is trained using physical governing equations and limited simulation or experimental data, new scenarios can be handled through lightweight adaptation or fine-tuning rather than full re-simulation.

Main Functions

1. **Physics-Embedded Learning Engine**
   - Direct incorporation of governing equations (PDEs/ODEs), boundary conditions, and conservation laws into neural network training.
   - Supports coupled multiphysics problems such as structure–fluid, thermal–fluid, and reaction–diffusion systems.
2. **Plug-in–Based Modular Architecture**
   - Independent modules for physics models, boundary/initial conditions, material constitutive and degradation models, and multiphysics coupling.
   - Enables flexible configuration and rapid deployment for different industrial simulation scenarios.
3. **Simulation Acceleration and Fast Re-Solving**
   - Provides rapid re-analysis for changed geometries or operating conditions through fine-tuning or adaptive learning.
   - Significantly reduces computational time compared to full-scale numerical simulations.
4. **Hybrid Integration with Existing Solvers**
   - Operates as a surrogate or co-simulation module compatible with commercial CAE/CFD tools.
   - Allows partial replacement of computational bottlenecks while preserving validated legacy solvers.
5. **Verification, Reliability, and Uncertainty Support**
   - Physics-consistency checks and residual-based error monitoring.
   - Supports uncertainty quantification to ensure trustworthy predictions for engineering decision-making.

Potential Uses and Applications

- **Automotive and Transportation**: Rapid evaluation of vehicle crashworthiness, structural response, and design optimization.
- **Manufacturing and Smart Engineering**: Accelerated virtual testing, process optimization, and digital engineering workflows.
- **Energy and Environment**: Multiphysics analysis for thermal systems, material degradation, and long-term performance prediction.
- **Digital Twin and Real-Time Simulation**: Integration into digital twin platforms for state estimation, predictive maintenance, and operational optimization.
- **R&D and Engineering Services**: High-throughput design exploration and reduced simulation cost in early-stage product development.

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Rationale for Joint R&D Project	The rapid increase in complexity of modern engineering systems—such as vehicles, energy systems, and advanced manufacturing processes—has made conventional multiphysics simulation increasingly computationally expensive and time-consuming. Industries now require fast, reliable, and repeatable simulation capabilities to support design iteration, optimization, and real-time decision-making. While Physics-Informed Neural Networks (PINNs) have emerged as a promising approach to accelerate scientific computing [2-4], their practical deployment in industrial-grade multiphysics problems remains limited due to challenges in robustness, scalability, validation, and integration with legacy simulation workflows. This creates a strong need for a joint R&D effort that combines complementary expertise across theory, industrial practice, and software implementation. **Need for the Joint R&D Project** No single organization can independently address the full technical and practical scope required to develop a universal PINN-based multiphysics simulation platform. Prof. Kwon at the University of Toronto possesses strong expertise in physics-based modeling, numerical simulation, and advanced PINN methodologies, but has limited access to large-scale, real-world industrial multiphysics datasets and operational environments. Conversely, the Korean SME owns and operates practical industrial simulation problems—such as crashworthiness analysis, CFD-based thermal systems, and material degradation modeling—and has direct experience with industrial users and workflows. However, the Korean SME faces constraints in independently developing advanced physics-informed AI architectures with sufficient theoretical rigor and generality. A joint R&D project is therefore essential to bridge advanced PINN research with deployable, user-ready industrial software. **Rationale for Collaboration** This collaboration enables the systematic integration of complementary strengths. Prof. Kwon at the University of Toronto leads the development of the core PINN algorithms, including governing-equation formulation, network architecture design, training stabilization strategies, and physics-consistency validation, ensuring scientific soundness and extensibility. The Korean SME contributes representative industrial use cases, high-fidelity simulation and operational data, and performance benchmarks necessary for model validation and tuning. In addition, the Korean SME is responsible for developing the user interface (UI) as well as pre-processing and post-processing modules, enabling practical data preparation, visualization, and result interpretation. This division of roles ensures that the developed technology is not only scientifically advanced but also accessible and usable by industrial engineers. **Strategic Importance of Joint Development** Joint R&D is strategically important to reduce both technical and commercialization risks. By co-developing the technology across multiple industrial application scenarios, the project avoids domain-

	specific overfitting and ensures that the resulting Universal PINN platform is broadly applicable. The involvement of the Korean SME in UI and pre-/post-processing development accelerates technology transfer, lowers user adoption barriers, and supports rapid deployment in industrial environments. Shared development and joint validation shorten the transition from research to commercialization, while jointly generated intellectual property and implementation know-how strengthen long-term competitiveness. Furthermore, collaboration between Prof. Kwon at the University of Toronto and the Korean SME enables the establishment of practical standards for physics-consistent AI-based simulation, which is increasingly required for industrial trust and future certification.
Key Words	*1. Physics-Informed Neural Networks (PINN)* *2. Multiphysics Simulation Acceleration* *3. AI-Based Scientific Computing* *4. Plug-in–Based Modular Software Architecture* *5. Digital Engineering and Virtual Modeling*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (8)
R&D Objectives	The objective of this R&D project is to develop a Universal Physics-Informed Neural Network (PINN)–based multiphysics simulation software platform that can replace or augment conventional numerical simulation tools by significantly reducing computational cost while maintaining physical consistency and industrial reliability. To achieve this overarching goal, the project focuses on the following specific technology development objectives: **1. Development of Core PINN Algorithms for Multiphysics Problems** • Design and implement robust PINN architectures that embed governing physical equations (PDEs/ODEs), boundary conditions, and conservation laws. • Support coupled multiphysics problems, including structural dynamics, fluid flow, thermal transport, and material degradation phenomena. **2. Implementation** • Develop a flexible plug-in framework that modularizes physics models, boundary/initial conditions, constitutive laws, and multiphysics coupling strategies. • Enable rapid configuration and reuse of PINN modules across different industrial simulation scenarios. **3. Fast Re-Solving and Adaptation Technology** • Develop fine-tuning and adaptive learning techniques that allow rapid re-analysis under changed geometries, materials, or operating conditions without full re-simulation. • Achieve significant acceleration compared to conventional CAE/CFD workflows. **4. Hybrid Integration with Existing Simulation Tools** • Develop interfaces that allow the PINN platform to operate as a surrogate or co-simulation module alongside commercial solvers. • Enable partial replacement of computational bottlenecks while preserving validated legacy simulation pipelines. **5. User-Oriented Software Components for Industrial Deployment** • Develop user interface (UI) modules and pre-processing and post-processing tools to support data preparation, model configuration, visualization, and result interpretation. • Ensure accessibility and usability for industrial engineers without deep AI expertise. **6. Verification, Reliability, and Uncertainty Handling Technologies** • Implement physics-consistency checks, residual-based error monitoring, and uncertainty quantification methods. • Provide trustworthy simulation outputs suitable for engineering decision-making and industrial validation.			
R&D Target Specifications	The R&D project aims to achieve the following quantitative and verifiable performance targets to demonstrate the effectiveness, reliability, and industrial applicability of the proposed Universal PINN–based multiphysics simulation platform. **1. Simulation Speed and Efficiency** • Simulation Time Reduction: Achieve at least 10×–50× reduction in computation time compared to conventional full-scale multiphysics simulations for representative industrial problems. • Fast Re-Solving Capability: Enable re-analysis under modified boundary conditions or geometries within minutes, compared to hours or days required by traditional solvers. **2. Prediction Accuracy and Physical Consistency**			

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	• **Solution Accuracy:** Achieve ≤ 5% relative error in key physical quantities (e.g., displacement, stress, temperature, velocity) compared to high-fidelity numerical simulation results. • Physics Residual Compliance: Maintain normalized governing-equation residuals below 10^{-3} across the computational domain. **3. Multiphysics Coupling Capability** • **Supported Physics Domains:** Successfully demonstrate coupled simulations involving at least two or more interacting physics domains (e.g., structure–thermal, fluid–thermal, diffusion–reaction). • **Stability under Coupling:** Achieve stable convergence for coupled problems with no divergence across ≥ 95% of tested scenarios. **4. Adaptability and Reusability** • **Fine-Tuning Data Efficiency:** Adapt to new operating conditions or geometries using ≤ 10% of the data required for initial model training. • **Model Reusability:** Enable reuse of trained PINN modules across at least three different industrial use cases without architecture redesign. **5. Software Usability and Integration** • **User Interface Responsiveness:** Complete model setup and result visualization within < 5 minutes using the developed UI and pre-/post-processing tools. • **Interoperability:** Support data exchange with at least two commercial CAE/CFD tools through standardized input/output formats. **6. System Maturity and Validation** • **Technology Readiness Advancement:** Advance the technology from TRL 3 (proof-of-concept) at project start to TRL 6 (prototype demonstrated in a relevant industrial environment) at project completion. • **Industrial Validation:** Validate the platform using minimum three representative industrial benchmark problems provided by the Korean SME.
R&D Plan	**1. Research Approach and Overall Strategy** The project adopts a hybrid AI–physics approach, in which governing physical equations are embedded into neural network models to ensure physical consistency, generalizability, and robustness. Rather than pursuing domain-specific PINN solutions, the research focuses on developing a plug-in–based Universal PINN architecture that can be flexibly configured for multiple multiphysics applications. The research strategy consists of: • Developing **core PINN algorithms and stabilization techniques** for multiphysics problems. • Modularizing physics, coupling, and adaptation mechanisms into reusable plug-ins. • Integrating the PINN core with **user-oriented software components** (UI, pre-/post-processing). • Validating the platform using representative industrial use cases provided by the Korean SME. **2. Key Tasks and Methodologies** **Task 1: Core PINN Algorithm and Physics Module Development** **Methodology** • Formulation of governing equations (PDEs/ODEs) for target physics domains. • Design of PINN architectures incorporating boundary/initial conditions and conservation laws. • Implementation of training stabilization techniques (adaptive loss weighting, domain decomposition, residual scaling).

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Output

- Verified PINN solvers for single-physics and basic coupled-physics problems.

Task 2: Multiphysics Coupling and Plug-in Architecture

Methodology

- Development of modular plug-ins for physics models, coupling interfaces, and constitutive/degeneration laws.
- Implementation of standardized APIs for plug-and-play configuration.
- Verification of stable convergence for coupled multiphysics cases.

Output

- Universal PINN core with configurable multiphysics plug-ins.

Task 3: Fast Re-Solving and Adaptation Techniques

Methodology

- Development of fine-tuning and transfer-learning strategies for rapid adaptation to new geometries and conditions.
- Data-efficient training using limited simulation or experimental datasets.
- Performance benchmarking against full numerical simulations.

Output

- Demonstrated 10×–50× simulation speed-up with controlled accuracy.

Task 4: Software Integration, UI, and Pre-/Post-Processing

Methodology

- Development of intuitive user interface (UI) for model setup and execution.
- Implementation of pre-processing tools for geometry, boundary condition, and data preparation.
- Development of post-processing and visualization tools for result interpretation and comparison.

Output

- User-operable prototype software platform suitable for industrial engineers.

Task 5: Industrial Validation and Performance Evaluation

Methodology

- Application of the platform to representative industrial benchmark problems (e.g., crashworthiness, thermal-fluid systems, material degradation).
- Quantitative evaluation of accuracy, speed, robustness, and usability.
- Iterative refinement based on validation feedback.

Output

- Validated Universal PINN prototype demonstrated in a relevant industrial environment.

3. Expected Milestones and Timeline

Phase	Period	Key Milestones
Phase 1	Year 1	Core PINN algorithms developed; single-physics validation completed (TRL 3→4)
Phase 2	Year 2	Multiphysics coupling and plug-in architecture implemented; fast re-solving demonstrated 4→6)
Phase 3	Year 3	UI, pre-/post-processing integrated; industrial validation completed; prototype demons (TRL 6→8)

4. Expected Outcomes

- A **Universal PINN-based multiphysics simulation software platform** with modular, plug-in architecture.
- Demonstrated acceleration and reliability compared to conventional simulation workflows.
- A deployable prototype suitable for industrial use, establishing a foundation for commercialization and further technology expansion.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute, led by Prof. Kwon at the University of Toronto, will play a central role in developing the scientific and algorithmic foundations of the proposed Universal PINN–based multiphysics simulation platform. The institute will be responsible for ensuring the technical rigor, generality, and reliability of the physics-informed AI technologies developed through this joint R&D project. **1. Technical Contributions** • Lead the development of core Physics-Informed Neural Network (PINN) algorithms, including network architectures that embed governing equations, boundary conditions, and conservation laws. • Formulate and implement multiphysics coupling strategies for structural, thermal, fluid, and material degradation problems. • Develop training stabilization and convergence enhancement techniques to improve robustness and scalability for industrial-scale simulations. **2. Research Scope and Methodology** • Conduct fundamental and applied research on physics-based modeling and numerical methods relevant to PINN-based simulation. • Design data-efficient learning and fine-tuning strategies to enable rapid adaptation to new geometries and operating conditions. • Establish verification and validation methodologies, including physics-consistency checks and error analysis. **3. Resources to Be Provided** • Provide expert research personnel, including faculty leadership, postdoctoral researchers, and graduate students with expertise in PINN, scientific computing, and multiphysics modeling. • Utilize advanced computational resources (e.g., GPU-enabled computing environments) for model development, training, and testing. • Supply research infrastructure and software tools necessary for algorithm development and benchmarking. **4. Specific Tasks and Deliverables** • Develop and deliver the core PINN engine and multiphysics algorithm modules to be integrated into the software platform. • Collaborate with the Korean SME to define standardized interfaces between the PINN core and user interface, pre-processing, and post-processing components. • Provide validated benchmark results and technical documentation to support industrial testing and deployment. • Contribute to joint intellectual property generation, academic dissemination, and knowledge transfer activities. **5. Collaboration and Project Management Support** • Participate in regular technical meetings and joint design reviews to ensure alignment across partners. • Support integration testing and performance evaluation in industrial environments. • Provide technical guidance to mitigate risks related to model accuracy, stability, and generalization.
Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in translating the core PINN technologies into an industrially deployable software platform. The SME will lead the practical development, validation, and user-facing implementation of the proposed Universal PINN–based multiphysics simulation system, ensuring its applicability to real-world industrial environments. **1. Technical Contributions** • Provide domain-specific engineering expertise in industrial multiphysics simulation problems, including crashworthiness analysis, CFD-based thermal systems, and material degradation modeling.

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- Define practical requirements, constraints, and performance criteria based on real industrial workflows and user needs.
- Contribute to the design of system architecture to ensure compatibility with existing industrial simulation pipelines.

2. Development Tasks

- Lead the development of user interface (UI) components for model configuration, execution control, and result visualization.
- Develop pre-processing modules for geometry handling, boundary condition setup, data preparation, and format conversion.
- Develop post-processing modules for visualization, comparison with reference simulations, performance reporting, and result interpretation.
- Integrate the PINN core modules provided by the research institute into a cohesive software prototype.

3. Testing and Validation Activities

- Provide representative industrial benchmark problems and high-fidelity simulation datasets for testing and validation.
- Conduct functional testing, performance benchmarking, and usability evaluation in industrial-like environments.
- Collaborate with the research institute to assess accuracy, speed, robustness, and reliability against conventional simulation tools.

4. Resource Commitments

- Assign dedicated software engineers and domain specialists to support platform development, integration, and testing.
- Provide computing and software resources required for system integration, UI development, and validation.
- Support pilot deployment and demonstration activities at the industrial level.

5. Collaboration and Commercialization Efforts

- Participate in joint technical meetings, design reviews, and iterative development cycles.
- Support technology transfer, documentation, and user training activities.
- Contribute to joint intellectual property generation and commercialization planning, including market feedback and deployment strategies.

In summary, the Korean SME partner will serve as the industrial implementation and validation lead, ensuring that the Universal PINN-based multiphysics simulation technology evolves from a research outcome into a practical, user-ready, and commercially viable software solution through close collaboration with the research institute.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver significant technological, industrial, and economic value by transforming Physics-Informed Neural Networks (PINNs) into a deployable, industrial-grade multiphysics simulation software platform. Through close collaboration between **Prof. Kwon at the University of Toronto** and the **Korean SME**, the project will bridge advanced physics-based AI research with real-world industrial applications. **1. Technological Achievements** • **Development of a Universal PINN Platform:** Establishment of a modular, plug-in–based PINN software architecture capable of handling a wide range of coupled multiphysics problems, including structural dynamics, thermal-fluid systems, and material degradation. • **Simulation Acceleration with Physical Reliability:** Demonstration of 10×–50× acceleration in simulation time compared to conventional numerical solvers, while maintaining prediction accuracy within industrially acceptable error bounds. • **Fast Re-Solving and Reusability:** Enable rapid re-analysis for changed geometries and operating conditions through fine-tuning and adaptive learning, significantly improving design iteration efficiency. • **Industrial-Grade Software Prototype:** Delivery of a validated software prototype integrating the PINN core, user interface, and pre-/post-processing tools, achieving TRL 6 by project completion. **2. Commercialization Potential** • **New Software Product and Service Opportunities:** The developed platform can be commercialized as a standalone simulation software, an add-on module to existing CAE/CFD tools, or a cloud-based simulation service. • **Reduced Time-to-Market for Industrial Products:** By shortening simulation cycles, the technology enables faster design optimization and virtual testing, directly contributing to reduced development costs and time-to-market. • **Scalability Across Multiple Industries:** The plug-in architecture allows rapid expansion into automotive, manufacturing, energy, and environmental sectors without fundamental redesign. • **Joint IP and Long-Term Competitiveness:** Jointly generated intellectual property and know-how strengthen the global competitiveness of the Korean SME and create opportunities for follow-on R&D and licensing. **3. Industrial and Economic Impact** • **Enhancement of Industrial Digital Engineering Capability:** Adoption of physics-consistent AI-based simulation tools advances the digital transformation of industrial engineering workflows. • **Cost Reduction and Productivity Gains:** Reduced reliance on repeated full-scale simulations lowers computational and engineering costs while improving productivity. • **Global Market Competitiveness:** The collaboration positions the Korean SME to enter high-value global markets for advanced engineering software and digital twin solutions. • **Foundation for Future Innovation:** The project establishes a foundation for next-generation digital twins, real-time simulation, and AI-assisted decision support systems, contributing to long-term industrial innovation and economic growth.

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6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization